

Explaining the Rationale of Deep Learning Glaucoma Decisions with Adversarial Examples

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Purpose: To illustrate what is inside the so-called black box of deep learning models (DLMs) so that clinicians can have greater confidence in the conclusions of artificial intelligence by evaluating adversarial explanation on its ability to explain the rationale of DLM decisions for glaucoma and glaucoma-related findings. Adversarial explanation generates adversarial examples (AEs), or images that have been changed to gain or lose pathologic characteristic-specific traits, to explain the DLM's rationale.

Design: Evaluation of explanation methods for DLMs.

Participants: Health screening participants (n = 1653) at the Seoul National University Hospital Health Promotion Center, Seoul, Republic of Korea.

Methods: We trained DLMs for referable glaucoma (RG), increased cup-to-disc ratio (ICDR), disc rim narrowing (DRN), and retinal nerve fiber layer defect (RNFLD) using 6430 retinal fundus images. Surveys consisting of explanations using AE and gradient-weighted class activation mapping (GradCAM), a conventional heatmap-based explanation method, were generated for 400 pathologic and healthy patient eyes. For each method, board-trained glaucoma specialists rated location explainability, the ability to pinpoint decision-relevant areas in the image, and rationale explainability, the ability to inform the user on the model's reasoning for the decision based on pathologic features. Scores were compared by paired Wilcoxon signed-rank test.

Main Outcome Measures: Area under the receiver operating characteristic curve (AUC), sensitivities, and specificities of DLMs; visualization of clinical pathologic changes of AEs; and survey scores for locational and rationale explainability.

Results: The AUCs were 0.90, 0.99, 0.95, and 0.79 and sensitivities were 0.79, 1.00, 0.82, and 0.55 at 0.90 specificity for RG, ICDR, DRN, and RNFLD DLMs, respectively. Generated AEs showed valid clinical feature changes, and survey results for location explainability were 3.94 ± 1.33 and 2.55 ± 1.24 using AEs and GradCAMs, respectively, of a possible maximum score of 5 points. The scores for rationale explainability were 3.97 ± 1.31 and 2.10 ± 1.25 for AEs and GradCAM, respectively. Adversarial example provided significantly better explainability than GradCAM.

Conclusions: Adversarial explanation increased the explainability over GradCAM, a conventional heatmap-based explanation method. Adversarial explanation may help medical professionals understand more clearly the rationale of DLMs when using them for clinical decisions. *Ophthalmology* 2021;128:78-88 © 2020 by the American Academy of Ophthalmology



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Deep learning models (DLMs) can detect glaucoma using retinal fundus images.^{1,2} However, the so-called black box nature of DLMs³ and the lack of valid explanations after DLM decisions limit the application and acceptance of these models in the medical field. Currently, the explainability of DLMs using retinal fundus imaging is limited to highlighting pixel areas of importance in the form of heatmaps using gradient-based saliency techniques^{4,5} or occlusion testing.⁶ In the context of glaucoma-related findings, heatmaps can identify the

location of an increased cup-to-disc ratio (ICDR) or disc rim narrowing (DRN)⁷ but cannot explain the DLM's rationale or the understanding of the pathologic characteristic-specific feature.

Understanding DLM decisions is important for clinical decision support,³ and the identification of important regions alone may be insufficient for understanding the model's rationale or assisting clinical decisions in the real world. Highlights of important regions are implicit, and users first must assume that the model's highlight

identifies the pathologic features by cause and not coincidence nor identification of other correlating features of the same region. Furthermore, determining the model's understanding using heatmaps alone assumes the user's ability to infer pathologic features from location, which may not always be the case with varying user expertise or clinical settings. Thus, location maps alone may fail to support clinical decisions by contributing no more than what the clinician can identify already. However, an explanation method that explicitly shows the model's understanding of pathologic characteristics with examples of added or removed pathologic features could provide a basis for reasonable and educational interactions among the clinician, the patient, and artificial intelligence and thus may enhance interaction, experience, and decision making.

Adversarial explanation is a method that uses the prediction DLM to make counterexamples as a means to explain the DLM's understanding of pathologic characteristic-specific features.⁸ For example, a valid adversarial explanation will consist of a negative adversarial example (AE) with decreased cup-to-disc ratio or a positive AE with increased cup-to-disc ratio given a retinal fundus image and a DLM for ICDR. Previously, AEs provided valid explanations for nonmedical images⁸; however, whether they provide valid explanations for retinal fundus images and whether they significantly increase the explainability of a model over current methods has not been reported. Thus, the clinical validation of adversarial explanation as a method for explaining DLMs for glaucoma and glaucoma-related findings is warranted.

Methods

Study Population and Image Data

Data at the Health Promotion Center, Seoul National University Hospital,⁹ from 2005 through 2016 were used for this study. Patients were anonymized before analysis was performed, and the need for patient consent was waived by the institutional review board at Seoul National University Hospital (institutional review board identifier, H-1703-044-837). The human ethics committees at Seoul National University Hospital approved the study. This work adhered to the tenets of the Declaration of Helsinki.

Annotation of Findings and Diagnosis

To minimize interobserver variability, each image was inspected by 3 physicians from a team of 9 board-trained ophthalmologists. Following International Society of Geographic and Epidemiologic Ophthalmology (ISGEO)¹⁰ guidelines and other well-designed studies,^{2,10,11} referable glaucoma (RG) was defined as having a vertical cup-to-disc ratio of 0.7 or more, DRN of 0.1 disc diameter or less, disc hemorrhage, or retinal nerve fiber layer defect (RNFLD). Respective findings for ICDR, DRN, and RNFLD were also annotated. Table S1 (available at www.aaojournal.org) shows the details of the criteria of glaucoma diagnosis, which was classified into normal, glaucoma suspect, and glaucoma most likely, based on the criteria of previous population studies.^{2,10,11} Referable glaucoma was defined as either glaucoma suspect or glaucoma most likely.

Fundus images were obtained using a digital nonmydriatic fundus camera (CR-2; Canon, Inc, Tokyo, Japan). Images were focused carefully on the posterior pole of the retina (macula centered) at a 45° field of view using the built-in split-line focusing device. Image quality was assessed based on field definition. Adequate field definition was confirmed only when (1) the entire macula and optic disc were visible and (2) both main temporal arcades were completely visible.¹² Retinal photographs of poor image clarity and poor field definition were regarded as nongradable. If any reviewer deemed a photograph as nongradable, it was excluded from the study.

Information on the age at visit, gender, diabetes medication, hypertension medication, and historical images were given for the most informed evaluation. We developed the interface program for annotation (Fig S1, available at www.aaojournal.org), and identical, color-calibrated monitors were distributed for annotation.

Development of Deep Learning Models

Patients were randomized into training, validation, and testing sets. To ensure that each subset contained a similar proportion of pathologic images, patient-randomized sampling was carried out until the proportion of pathologic images had at least the prevalence of the least prevalent finding, namely that of RNFLD (approximately 2.5%). The training set was used for model training, the validation set was used to determine when to end training, and the testing set was used for performance measurements and explanations. The resulting training, validation, and testing sets consisted of 4867, 544, and 605 images, respectively.

Deep learning models were trained separately to predict RG and each glaucoma-related finding: ICDR, DRN, and RNFLD. To resolve class imbalance, we upsampled pathologic images in the training set such that 1 in 3 images showed pathologic findings. Images were resized to 224 × 224 pixels during training and testing.

A ResNet-50 was used as the backbone of our models,¹³ which outputs 2 numbers representing pathologic or nonpathologic and that add up to 1. Training was carried out for 100 epochs, using transfer learning from a model trained on the ImageNet database,¹⁴ saving the model with the best validation set cross-entropy loss or area under the receiver operating characteristic curve (AUC). During training, we used random crop, random vertical flip, MixUp,¹⁵ and random horizontal flip. Batches comprised half AEs and half original images, otherwise known as half-half training.¹⁶

Explanation Using Adversarial Examples

We produced adversarial examples (AEs) using adversarial explanation Woods et al⁸ use backpropagation from a DLM iteratively to alter the noise, or the adversarial noise, given to an image that maximizes the model's prediction for a predetermined category. They introduced $g_{\text{explain}+/-}$, in which the shape of the noise was optimized to reveal salient, perceptually different examples. Simply put, adversarial explanation is a method of "repainting" the image until the model regards it as either completely pathologic or completely normal. We denoted AEs that were encouraged to be pathologic as positive AEs and those that were encouraged to be normal as negative AEs. Adversarial examples were made by 100 iterations of backpropagation.

For a comparison with a conventional heatmap-based explanation method, we produced saliency heatmaps using gradient-weighted class activation mapping (GradCAM).⁴ To compare GradCAM with AE directly, saliency heatmaps were also generated for the adversarial explanations by taking the absolute value of the pixel values in the adversarial noise. This is akin to

identifying the areas of the AEs where the most change occurred. Because GradCAM heatmaps are small, usually 7×7 pixels depending on the DLM, we used bicubic interpolation when overlaying on the original image.

Expert Survey of Deep Learning Explanation Methods

A survey was conducted by 3 board-trained ophthalmologists to compare the explainability of AEs and GradCAM. From the testing set, we sampled 400 patient eyes, which consisted of 100 images each for RG, ICDR, DRN, and RNFLD models. Because of the small prevalence of pathologic images, all pathologic images in the testing set were included in the survey and normal images were sampled randomly from the testing set. For each image, we produced an explanation using AEs and an explanation using GradCAM (Fig S2, available at www.aaojournal.org). For each explanation, 2 questions were asked: “How well does this explanation method explain the location of the normal or pathologic finding?” and “How well does this explanation method explain the rationale of the normal or pathologic finding?” Each question was given a score between 1 and 5. Because 1 AE explanation and 1 GradCAM explanation were shown sequentially for the same image, the difference in scores was compared in pairs to determine whether 1 explanation produced better explainability than the other with respect to location or rationale.

Statistical Methods

The associations between findings and diagnoses were determined by the chi-square test. The Fleiss κ for 3 raters was used to calculate interobserver agreement. Survey results were analyzed using the paired Wilcoxon signed-rank test, which tests the significance of the difference of scores without the need for any assumption in the distribution of scores. Alpha values of less than 0.05 were considered statistically significant. For deep learning, we used Fastai v1 (fast.ai Inc, CA) with Pytorch (Facebook Inc) backend on Python 3.6 (Python Software Foundation).

Results

From 6430 images that were annotated 3-fold, 417 were excluded for poor image quality. Our final dataset consisted of 1653 patients comprising 3091 distinct visits and 6013 images. The prevalences of RG, ICDR, DRN, and RNFLD were 9.6%, 4.4%, 6.0%, and 2.6%, respectively, and interobserver agreement κ values were 0.461, 0.499, 0.367, and 0.188, respectively (Table 1). Overall, 54.6% of images were from men who had a mean age of 55.7 ± 9.66 years at the time of the examination (Table 1). Patient-randomized training, validation, and testing subsets consisted of at least 15 or more than 2.5% of each pathologic feature, except for disc hemorrhage, which was not used for training (Table 1). A significant association was found between RG and pathologic findings (Table S2, available at www.aaojournal.org). The RG, ICDR, DRN, and RNFLD models had AUCs of 0.899, 0.986, 0.950, and 0.790, respectively, for the testing subset. When stratified by greater severity of glaucoma, namely most likely glaucoma diagnosis, models for RG, ICDR, DRN, and RNFLD showed AUCs of 0.989, 0.991, 0.987, and 0.900, respectively. The receiver operating characteristics stratified by severity of glaucoma are shown in Figure 1. Sensitivities were 0.79, 1.00, 0.82, and 0.55 for RG, ICDR, DRN, and RNFLD, respectively, at the preset specificity of 0.90.

Adversarial Example

Figure 2 shows adversarial explanations and GradCAM heatmaps for RG, RNFLD, ICDR, and DRN DLM decisions for selected images. Figure 2A depicts adversarial explanations and GradCAM heatmaps for the RG DLM on a positive sample. The positive GradCAM shows that RG is the result of the disc area and the negative GradCAM highlights nondisc regions. The negative AE for referable glaucoma ($AE_{\text{glaucoma}-}$) has decreased the cup-to-disc ratio and widened the superiotemporal disc rim, which the DLM predicts as negative for RG. The positive AE for referable glaucoma ($AE_{\text{glaucoma}+}$), which results in a higher prediction for glaucoma, increases cup-to-disc ratio and narrows the superiotemporal disc rim. Figure 2B depicts explanations for the RG DLM on a negative sample. The negative GradCAM highlights the fovea region and parts outside the field of view (FOV), and the positive GradCAM highlights several non-FOV areas. The $AE_{\text{glaucoma}+}$ adds an increased cup-to-disc ratio finding, which the original image lacks. Figure 2C depicts a sample with RNFLD. The positive GradCAM highlights regions outside the FOV, and the negative GradCAM highlights the region inferior to the disc. The negative AE for RNFLD ($AE_{\text{RNFLD}-}$) shows diminished RNFLD, and the positive AE for RNFLD ($AE_{\text{RNFLD}+}$) shows a darker defect. Adversarial noises show that the most changes have been made at the area of RNFLD. Figure 2D depicts a sample without RNFLD, for which GradCAM highlights regions outside the FOV. The $AE_{\text{RNFLD}-}$ does not make apparent changes to the original image, whereas the $AE_{\text{RNFLD}+}$ shows darkened nerve fiber regions temporally and superotemporally to the disc. Figure 2E, F depicts the ICDR-positive sample and ICDR-negative sample, respectively. Gradient-weighted class activation mapping correctly highlights the disc region. Relevant changes in cup-to-disc ratio are shown explicitly in the positive AE for ICDR ($AE_{\text{ICDR}+}$), which increases the cup-to-disc ratio, and the negative AE for ICDR ($AE_{\text{ICDR}-}$), which decreases the cup-to-disc ratio. Figure 2G depicts a positive DRN image. The location of the disc is identified correctly in the positive GradCAM for this DRN-positive sample. The widening and narrowing changes to superiotemporal disc rim width can be seen in the negative AE for DRN ($AE_{\text{DRN}-}$) and the positive AE for DRN ($AE_{\text{DRN}+}$), respectively. Figure 2H depicts a negative DRN image. The GradCAM heatmap fails to locate the disc. However, the $AE_{\text{DRN}+}$ shows a superiotemporal disc rim narrowing, which the original lacks.

Zoomed-in and annotated adversarial explanations are shown in Figure 3. Adversarial examples show valid clinical feature changes, including changes to peripapillary atrophy, RNFLD, cup-to-disc ratio, and DRN. Figure 3A shows an example with ICDR, for which the $AE_{\text{glaucoma}-}$ has removed peripapillary atrophy (white arrows) and reduced cup size (black arrow), whereas the $AE_{\text{glaucoma}+}$ has exaggerated the peripapillary border (white arrows) and increased cup size (black arrow). Figure 3B shows an example with only RNFLD, for which the $AE_{\text{glaucoma}-}$ has replaced the RNFLD with disconnected vessel-like patterns (white arrows), and the $AE_{\text{glaucoma}+}$ darkens and widens the RNFLD (black arrows). Figure 3C shows an example with ICDR and DRN, for which the $AE_{\text{ICDR}-}$ shows reduced cup size (white arrows), and the $AE_{\text{ICDR}+}$ shows increased cup size (black arrows) and added bright peaks near cup margins (black arrows). Figure 3D shows an example with DRN, for which the $AE_{\text{DRN}-}$ increases the superiotemporal rim width (black arrows), and the $AE_{\text{DRN}+}$ makes the disc rim narrower. Figure 3C, D appears defocused because it is zoomed in on sections of images of 224×224 pixels, a size requirement for input into the DLM. Figure 3E shows an example of RNFLD, for which the $AE_{\text{RNFLD}-}$ fills the RNFLD with discontinuous fiber-like patterns (green arrows), and the $AE_{\text{RNFLD}+}$ produces a defect with darker contrast (green arrow) and thins the bridging vessel (red arrow).

Table 1. Sample Characteristics, Associations between Findings and Diagnoses, Interobserver Agreement, and Deep Learning Model Results

Data*	Total	Pathologic Feature				Randomization Subsets		
		Referable Glaucoma [†]	Increased Cup-to-Disc Ratio [‡]	Disc Rim Narrowing [‡]	Retinal Nerve Fiber Layer Defect [‡]	Training Set	Validation Set	Testing Set
No. of images (%)	6013 (100)	579 (9.6)	267 (4.4)	358 (6)	157 (2.6)	4864 (80.9)	544 (9)	605 (10.1)
No. of patient visits (%)	3091 (100)	402 (13)	171 (5.5)	253 (8.2)	132 (4.3)	2504 (81)	276 (8.9)	311 (10.1)
No. of patients (%)	1653 (100)	223 (13.5)	82 (5)	138 (8.3)	76 (4.6)	1322 (80)	165 (10)	166 (10)
Characteristics								
Age (yrs), mean (SD)	55.7 (9.59)	55.8 (9.7)	56.1 (9.0)	56.2 (10.0)	56.0 (10.1)	55.76 (9.60)	54.23 (9.36)	56.18 (9.60)
Male gender, no. (%)	3,284 (54.6)	374 (64.6)	196 (73.4)	240 (67.0)	102 (65.0)	2657 (54.6)	312 (57.3)	315 (52.1)
Referable glaucoma, no. (%) [†]	579 (9.6)	579 (100)	265 (99.3)	354 (98.9)	153 (97.5)	468 (9.6)	53 (9.7)	58 (9.6)
Glaucoma suspect	472 (7.8)	472 (81.5)	187 (70.0)	255 (71.2)	72 (45.9)	388 (8.0)	43 (7.9)	41 (6.8)
Glaucoma most likely	107 (1.8)	107 (18.5)	78 (29.2)	99 (27.7)	81 (51.6)	80 (1.6)	10 (1.8)	17 (2.8)
Increased C/D ratio, no. (%) [†]	267 (4.4)	265 (45.8)	267 (100)	179 (50)	70 (44.6)	214 (4.4)	28 (5.1)	25 (4.1)
Disc rim narrowing, no. (%) [†]	358 (6.0)	354 (61.1)	179 (67)	358 (100)	102 (65)	288 (5.9)	36 (6.6)	34 (5.6)
RNFLD, no. (%) [†]	157 (2.6)	153 (26.4)	70 (26.2)	102 (28.5)	157 (100)	122 (2.5)	15 (2.8)	20 (3.3)
Interobserver agreement								
κ statistic [‡]		0.461	0.499	0.367	0.188			
Model performance among testing set, AUC (95% CI)								
Overall		0.899 (0.870–0.928)	0.986 (0.977–0.995)	0.950 (0.930–0.970)	0.790 (0.712–0.868)			
Among no glaucoma and GS subgroup		0.862 (0.820–0.904)	0.985 (0.974–0.996)	0.925 (0.889–0.961)	0.705 (0.574–0.836)			
Among no glaucoma and GM subgroup		0.989 (0.981–0.997)	0.991 (0.983–0.999)	0.987 (0.978–0.996)	0.900 (0.838–0.962)			

AUC = area under receiver operating characteristic curve; C/D = cup-to-disc; CI = confidence interval; GM = glaucoma most likely; GS = glaucoma suspect; RNFLD = retinal nerve fiber layer defect.

*Percentage of data using proportion of subset among total set (rows) and percentage of characteristics using proportions of images within the total set or subsets (columns).

[†]Significant associations ($P < 0.05$) between all combinations of glaucoma, increased C/D ratio, RNFLD, and disc rim narrowing using the chi-square test was observed and is shown in [Table S1](#).

[‡]Fleiss κ for 3 raters.

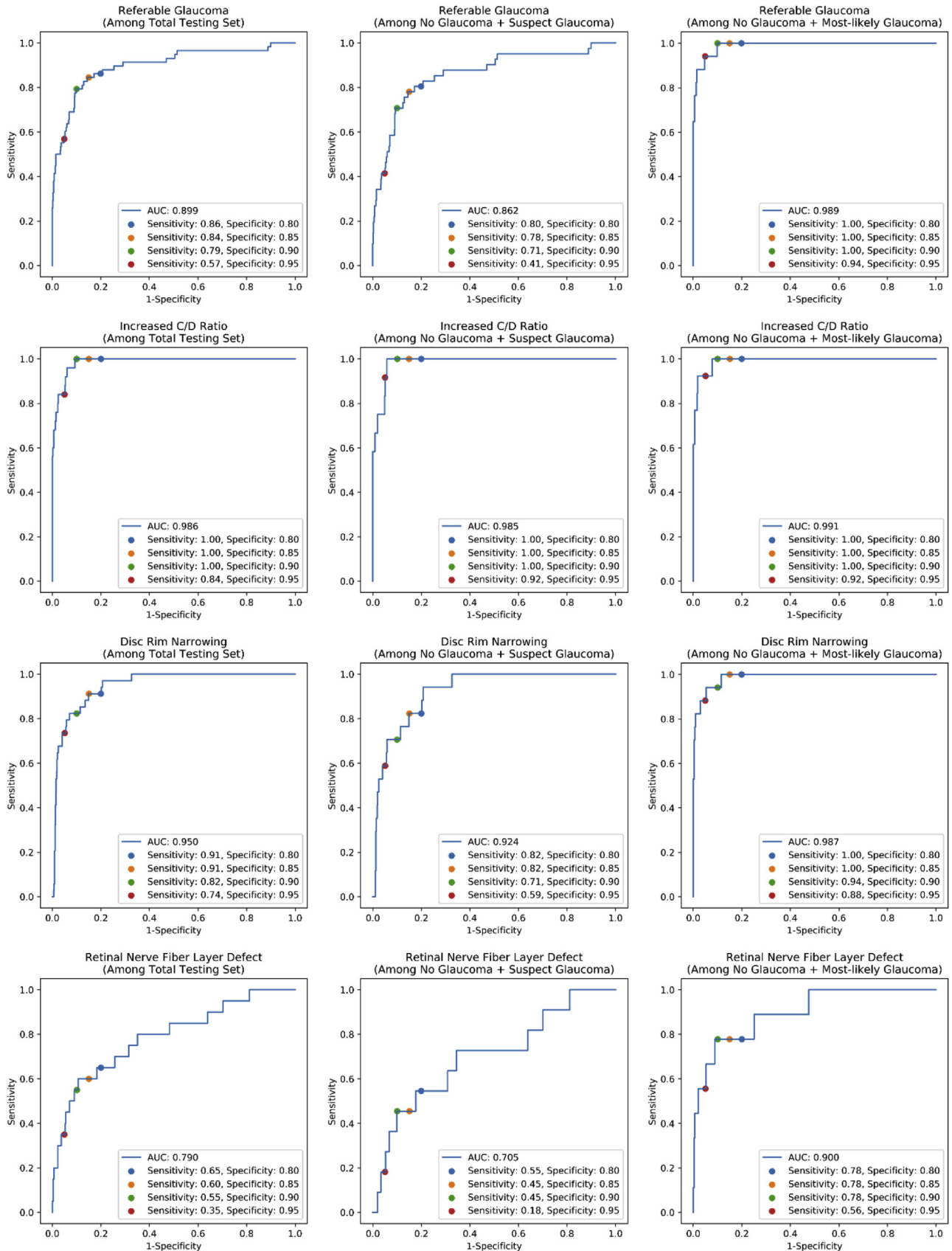


Figure 1. Receiver operating characteristic curves for referable glaucoma, increased cup-to-disc ratio, disc rim narrowing, and retinal nerve fiber layer defect stratified by severity of glaucoma. AUC = area under curve; C/D = cup-to-disc.

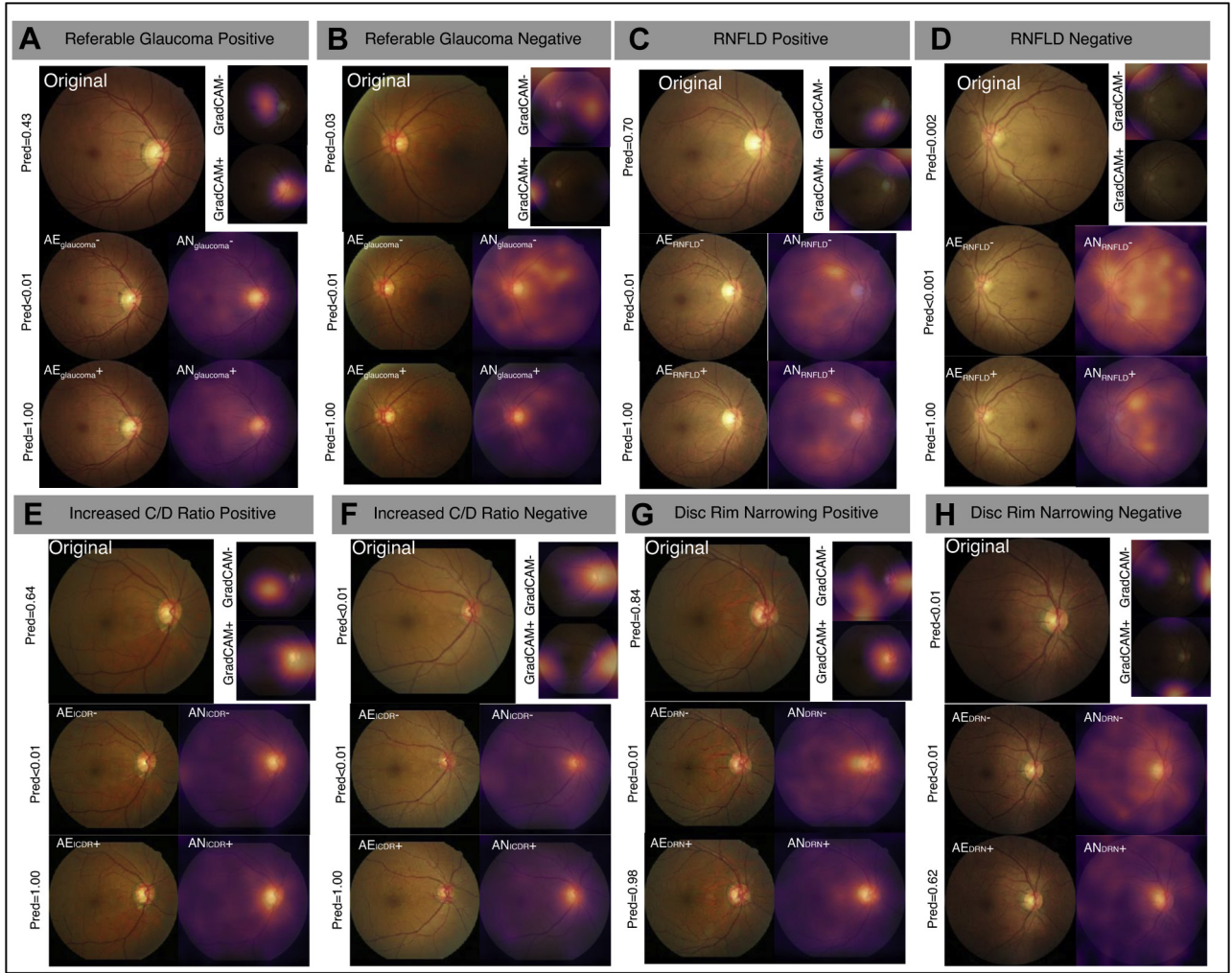


Figure 2. Adversarial explanations of deep learning models for referable glaucoma, retinal nerve fiber layer defect (RNFLD), increased cup-to-disc (C/D) ratio (ICDR), and disc rim narrowing (DRN). **A**, Adversarial explanation for a referable glaucoma positive image. **B**, Adversarial explanation for a referable glaucoma negative image. **C**, Adversarial explanation for an RNFLD positive image. **D**, Adversarial explanation for an RNFLD negative image. **E**, Adversarial explanation for an ICDR positive image. **F**, Adversarial explanation for an ICDR negative image. **G**, Adversarial explanation for a DRN positive image. **H**, Adversarial explanation for a DRN negative image. All prediction values indicate probability for positive (+) pathologic features. First row shows the original image and gradient-weighted class activation mapping (GradCAM) for positive and negative (−) pathologic features. The second row is the negative adversarial example (AE) with respective adversarial noise (AN). The third row is the positive AE with respective AN. Adversarial examples were produced using only the predictive deep learning model and the original sample image. Adversarial noise is shown superimposed on the original image. Pred = prediction.

Survey Results

Table 2 shows the survey results for location explainability (question 1), and Table 3 shows the survey results for rationale explainability (question 2). The average score for location explainability was 3.94 ± 1.33 and 2.55 ± 1.24 for AE and GradCAM, respectively, of a possible maximum score of 5 points. When rating location explainability, raters gave on average 1.39 points higher for AE than GradCAM ($P < 0.001$). The average score for rationale explainability was 3.97 ± 1.31 and 2.10 ± 1.25 for AE and GradCAM, respectively. When rating rationale explainability, raters on average gave 1.87 points higher for AE than GradCAM ($P < 0.001$). Adversarial example received higher scores for both location and rationale explanations among normal images (1.59 ± 1.56 and 2.03 ± 1.60 , respectively), more so than among pathologic images (1.00

± 1.45 and 1.58 ± 1.74 , respectively). Survey scores for RG, DRN, and ICDR resulted in favor of AE, with 1.13 ± 1.35 -, 1.91 ± 1.38 -, and 2.19 ± 1.23 -point differences for location explainability and 1.98 ± 1.52 -, 2.20 ± 1.42 -, and 2.69 ± 1.31 -point differences for rationale explainability. However, absolute grade scores were low for RNFLD at less than 3 points and resulted in minimal explainability differences between AE and GradCAM. For location explainability, GradCAM scored marginally higher than AE among RNFLD-positive images.

Discussion

To date, the implementation of explainable DLMs on glaucoma prediction using retinal fundus images using adversarial explanations has not been reported.

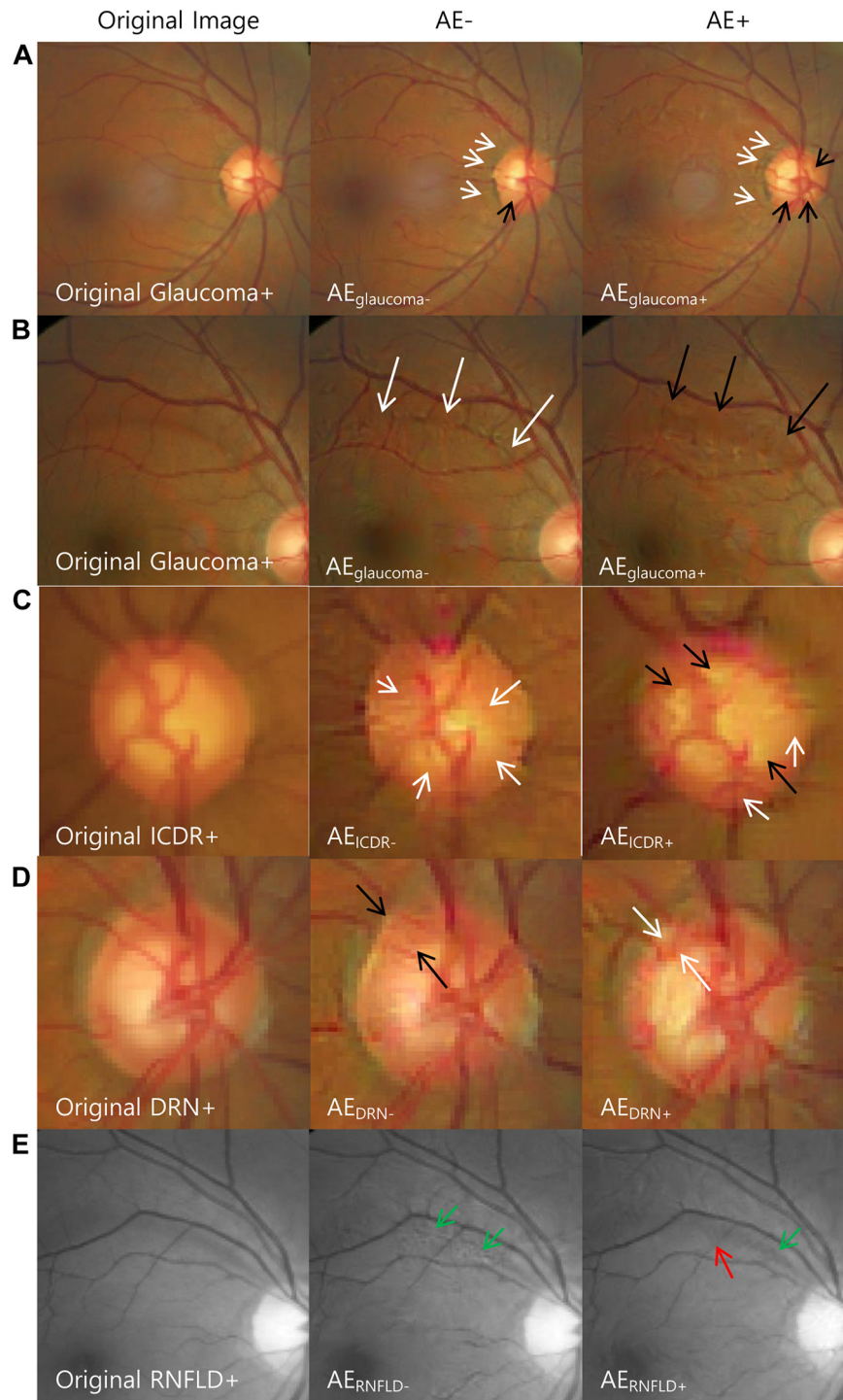


Figure 3. Examples of zoomed-in adversarial explanations for deep learning models for referable glaucoma, increased cup-to-disc ratio (ICDR), disc rim narrowing (DRN), and retinal nerve fiber layer defect (RNFLD). The first column is the original image, the second column is the negative (–) adversarial example (AE), and the third column is the positive (+) AE. **A**, Example with ICDR, for which the negative AE for referable glaucoma (AE_{glaucoma-}) has removed peripapillary atrophy (white arrows) and reduced cup size (black arrow), whereas the positive AE for referable glaucoma (AE_{glaucoma+}) has exaggerated the peripapillary atrophy (white arrows) and increased cup size (black arrow). **B**, Example with only RNFLD, for which AE_{glaucoma-} has replaced the RNFLD with disconnected vessel-like patterns (white arrows), and AE_{glaucoma+} darkens and widens the RNFLD (black arrows). **C**, Example with ICDR and DRN, for which the negative AE for ICDR (AE_{ICDR-}) has reduced cup size (white arrows), and the positive AE for ICDR (AE_{ICDR+}) has increased cup size (black arrows) and added bright peaks near cup margins (black arrows). **D**, Example with DRN, for which the negative AE for DRN (AE_{DRN-}) increases the superiotemporal rim width (black arrows), and the positive AE for DRN (AE_{DRN+}) makes the disc rim narrower. **(C)** and **(D)** appear defocused because they are zoomed-in sections of images of size of 224 × 224 pixels, a size requirement for input into the deep learning model. **(E)** shows an example of RNFLD, for which the negative AE for RNFLD (AE_{RNFLD-}) fills the RNFLD with discontinuous fiberlike patterns (green arrows), and the positive AE for RNFLD (AE_{RNFLD+}) produces a defect with darker contrast (green arrow) and thins the bridging vessel (red arrow).

Table 2. Comparison of Scores for “How Well Does This Explanation Method Explain the Location of the Normal or Pathologic Finding?” of Each Explanation Method

Models	Subgroups	No. of Images	Total No. of Surveys	Score for Adversarial Explanation (Mean $\pm$ Standard Deviation)	Score for Gradient-weighted Class Activation Mapping (Mean $\pm$ Standard Deviation)	Paired Score Difference (Mean $\pm$ Standard Deviation)	P Value*
All Models (referable glaucoma, DRN, ICDR, and RNFLD)							
	All	400	1200	3.94 $\pm$ 1.33	2.55 $\pm$ 1.24	1.39 $\pm$ 1.55	< 0.001
	Pathologic	137	411	4.14 $\pm$ 1.25	3.14 $\pm$ 1.20	1.00 $\pm$ 1.45	< 0.001
	Normal	263	789	3.83 $\pm$ 1.36	2.24 $\pm$ 1.14	1.59 $\pm$ 1.56	< 0.001
Referable glaucoma							
	All	100	300	3.99 $\pm$ 0.93	2.86 $\pm$ 1.17	1.13 $\pm$ 1.35	< 0.001
	Referable glaucoma positive	58	174	4.17 $\pm$ 0.88	2.91 $\pm$ 1.22	1.25 $\pm$ 1.40	< 0.001
	Referable glaucoma negative	42	126	3.75 $\pm$ 0.95	2.79 $\pm$ 1.11	0.96 $\pm$ 1.26	< 0.001
DRN							
	All	100	300	4.76 $\pm$ 0.56	2.85 $\pm$ 1.34	1.91 $\pm$ 1.38	< 0.001
	DRN positive	34	102	4.87 $\pm$ 0.41	3.77 $\pm$ 1.05	1.10 $\pm$ 1.16	< 0.001
	DRN negative	66	198	4.70 $\pm$ 0.61	2.37 $\pm$ 1.21	2.33 $\pm$ 1.29	< 0.001
ICDR							
	All	100	300	4.87 $\pm$ 0.47	2.68 $\pm$ 1.09	2.19 $\pm$ 1.23	< 0.001
	ICDR positive	25	75	4.84 $\pm$ 0.59	3.41 $\pm$ 0.87	1.43 $\pm$ 1.14	< 0.001
	ICDR negative	75	225	4.88 $\pm$ 0.42	2.44 $\pm$ 1.04	2.45 $\pm$ 1.16	< 0.001
RNFLD							
	All	100	300	2.12 $\pm$ 0.92	1.80 $\pm$ 1.03	0.32 $\pm$ 1.51	< 0.001
	RNFLD positive	20	60	1.93 $\pm$ 1.16	2.37 $\pm$ 1.15	-0.43 $\pm$ 1.58	0.049
	RNFLD negative	80	240	2.17 $\pm$ 0.84	1.66 $\pm$ 0.95	0.50 $\pm$ 1.44	< 0.001

DRN = disc rim narrowing; ICDR = increased cup-to-disc ratio; RNFLD = retinal nerve fiber layer defect.

*Wilcoxon signed-rank test.

Table 3. Comparison of Scores for “How Well Does This Explanation Method Explain the Rationale of the Normal or Pathologic Finding?” of Each Explanation Method

Models	Subgroups	No. of Images	Total No. of Surveys	Score for Adversarial Explanation (Mean $\pm$ Standard Deviation)	Score for Gradient-Weighted Class Activation Mapping (Mean $\pm$ Standard Deviation)	Paired Score Difference (Mean $\pm$ Standard Deviation)	P Value*
All models (referable glaucoma, DRN, ICDR, RNFLD)							
	All	400	1200	3.97 $\pm$ 1.31	2.10 $\pm$ 1.25	1.87 $\pm$ 1.66	< 0.001
	Pathologic	137	411	4.19 $\pm$ 1.24	2.61 $\pm$ 1.39	1.58 $\pm$ 1.74	< 0.001
	Normal	263	789	3.86 $\pm$ 1.33	1.84 $\pm$ 1.08	2.03 $\pm$ 1.60	< 0.001
Referable glaucoma							
	All	100	300	4.20 $\pm$ 0.94	2.22 $\pm$ 1.25	1.98 $\pm$ 1.52	< 0.001
	Referable glaucoma positive	58	174	4.35 $\pm$ 0.89	2.33 $\pm$ 1.31	2.02 $\pm$ 1.58	< 0.001
	Referable glaucoma negative	42	126	4.00 $\pm$ 0.96	2.06 $\pm$ 1.15	1.94 $\pm$ 1.44	< 0.001
DRN							
	All	100	300	4.62 $\pm$ 0.76	2.42 $\pm$ 1.42	2.20 $\pm$ 1.42	< 0.001
	DRN positive	34	102	4.72 $\pm$ 0.67	3.10 $\pm$ 1.53	1.62 $\pm$ 1.46	< 0.001
	DRN negative	66	198	4.57 $\pm$ 0.80	2.07 $\pm$ 1.22	2.51 $\pm$ 1.30	< 0.001
ICDR							
	All	100	300	4.80 $\pm$ 0.63	2.11 $\pm$ 1.16	2.69 $\pm$ 1.31	< 0.001
	ICDR positive	25	75	4.73 $\pm$ 0.79	2.75 $\pm$ 1.36	1.99 $\pm$ 1.60	< 0.001
	ICDR negative	75	225	4.83 $\pm$ 0.57	1.90 $\pm$ 1.00	2.93 $\pm$ 1.11	< 0.001
RNFLD							
	All	100	300	2.27 $\pm$ 0.97	1.66 $\pm$ 1.01	0.61 $\pm$ 1.63	< 0.001
	RNFLD positive	20	60	2.15 $\pm$ 1.29	2.40 $\pm$ 1.18	-0.25 $\pm$ 1.63	0.213
	RNFLD negative	80	240	2.30 $\pm$ 0.88	1.47 $\pm$ 0.87	0.83 $\pm$ 1.56	< 0.001

DRN = disc rim narrowing; ICDR = increased cup-to-disc ratio; RNFLD = retinal nerve fiber layer defect.

*Wilcoxon signed-rank test.

Explainability in deep learning-based predictions have largely been focused on identifying the region of importance,^{4,5} using, for example, saliency maps.⁷ This can explain where specific pathologic features exist but cannot provide an explanation of the rationale for its decision. Adversarial explanations can provide model-derived positive and negative examples as a means to explain the model's rationale.⁸ Herein, we showed that explanations using AEs may enhance the location and rationale explainability of a model over traditional saliency techniques such as GradCAM. Furthermore, an analysis of select AEs showed that adversarial explanation can add or remove RG and glaucoma-related findings to explain DLM decisions.

In this study, trained DLMs achieved similar AUCs to previous studies^{1,2,7,17–21}; however, comparisons should consider the differences in severity of disease from differing study populations. Generally, clinic-based datasets have higher prevalences of and greater severity in cases compared with a screening-based dataset, and the difference in severity and population is known to affect AUC results. For example, a previous study reported an AUC of 0.945 for RG in a clinic-based validation set but 0.855 in an external screening-based validation set.¹ One study showed that the AUC was 0.89 for mild cases but 0.97 for moderate to severe cases of glaucoma.¹⁸ Most studies use clinic-based datasets to report AUCs for RG, which range from 0.89 to 0.986.^{2,17–20} These studies vary in the way data are collected and may not represent real-world populations because of variable inclusion and exclusion criteria of images. In community-based testing samples, previous work reported AUCs for RG between 0.855¹ and 0.942.²¹ Few studies developed and validated models for specific glaucoma-related findings.^{1,7} A study using a clinic-based dataset reported AUCs of 0.922, 0.946, and 0.778 for ICDR, DRN, and RNFLD,¹ respectively, and a study using a screening-based dataset reported AUCs of 0.982 and 0.983 for glaucomatous disc change and RNFLD, respectively.⁷ Our study, despite being based in a screening setting having lower prevalence and severity of disease, reported an AUC of 0.989 for RG and comparable AUCs for glaucoma-related findings. When stratified by disease severity, the AUC for our models improved when detecting findings among most likely glaucoma patients.

Large differences were found in the visualizations of what the model used by applying adversarial explanation and GradCAM strategies. These differences occurred because of how the explanations were made. With positive GradCAM, areas of the original image that contributed to a positive prediction were highlighted, whereas with positive AE, the example with added pathologic features was shown along with an adversarial noise-positive heatmap of areas with the most change. Thus, positive GradCAM did not highlight anything at all if the image was totally normal, as seen in Figure 2B, D, whereas positive adversarial noise highlighted the area that changed most in positive AE.

Survey results showed that adversarial explanations provided better location and rationale explainability over GradCAM. The reasons for these improvements are multifactorial. First, GradCAM had no fundamental logic for

explaining the rationale of the model's decision. The information contained in a saliency map is purely locational, highlighting the regions contributing to or against a decision. On the contrary, AE provided logic for explaining the rationale behind the model's decision, namely, an example with removed pathologic features (negative AE) and an example with added pathologic features (positive AE). Second, GradCAM explanations performed poorly when given normal images. Gradient-based saliency techniques have had trouble interpreting heatmaps for normal findings.^{22,23} Similarly, in our study, the survey results showed lower explainability for GradCAM among normal images. It is probable that for normal cases, the positive GradCAM heatmaps did not highlight the region of interest correctly because the image features did not make any contributions toward a pathologic prediction. However, AEs performed well for both pathologic and normal images because they were able to add pathologic characteristic-specific features to normal images to give a more definitive location.

Expert survey showed successful explanations for RG, ICDR, and DRN. The validation of adversarial explanation has potential applications such as deep learning-based medical devices with greater explainability, medical education with generated AEs, and predictive visualization of glaucoma progression.

Our study includes several strengths. First, our work validated a deep learning explanation method that has not been applied and validated for retinal fundus images and provided a guide for validating explanation methods for other medical imaging methods. Second, our dataset was annotated prospectively for deep learning purposes using a specific criterion for RG, self-developed annotation software, and unified color-calibrated monitors, which minimizes misclassification bias. Third, our survey assessed 2 different aspects of explainability, namely location and rationale, that previous studies have not addressed separately.³

Our data have limited diversity compared with those from other studies^{1,2} and did not validate on external datasets, which may limit the generalizability of our DLMs. However, the primary objective of our work was to validate the explainability of an explanation technique rather than the diagnostic performance of a DLM. Although our model may behave differently on external datasets, the adversarial explanation technique can be generalized and used for DLMs developed elsewhere.

Our model for RNFLD did not produce a high AUC (0.790) and resulted in only a small increase in explainability by adversarial explanations. Previous studies report similar¹ or higher⁷ AUC, and we think that RNFLD did not perform as well as other glaucoma parameters for the following reasons. First, the number of RNFLD training examples was small (122 images) compared with that of other studies (1168 images).⁷ It is widely believed that a large dataset is crucial in medical deep learning,²⁴ and the number of RNFLD images may have been too few for the DLM to be trained successfully. Second, the characteristics of the existing RNFLD images of our database may have been subtle and ambiguous because the study population was based on a health screening setting and not a glaucoma

clinic setting. The screening setting represents relatively earlier stages of glaucoma in which RNFLDs are smaller or harder to detect. This is evident in our analysis of interobserver agreement, where RNFLD showed lower agreement (0.19) compared with other glaucoma parameters (> 0.35). Third, the RNFLD model performed poorly when detecting less severe cases of disease. When tasked with detecting RNFLD without less severe, suspected glaucoma patients, the AUC for RNFLD increased to 0.900, whereas excluding more severe, most likely glaucoma patients, the AUC dropped to 0.705. Because our population consisted of health screening participants, the existence of less severe cases of disease resulted in lower performance. Finally, the lower performance of the RNFLD model likely resulted in lower explainability. Compared with all other models, RNFLD showed low explainability scores for both AE and

GradCAM. Although the differences between the 2 methods were insignificant or marginally better for GradCAM in positive RNFLD subgroups, the low absolute scores indicate the inability of the model to consistently produce accurate and valid explanations, rather than the outperformance of one method over the other. This warrants further work using more robust and accurate DLMs for RNFLD.

In conclusion, the expert survey showed significant improvements in location and rationale explainability of DLMs when using AE compared with GradCAM, and AE produced pathologic characteristic-specific feature changes to explain deep learning-based decisions for RG and glaucoma-related findings. Adversarial explanation may help medical professionals to understand DLM-based decisions better.

Footnotes and Disclosures

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Abbreviations and Acronyms:

AE = adversarial example; **AE_{DRN-}** = negative adversarial example for disc rim narrowing; **AE_{DRN+}** = positive adversarial example for disc rim narrowing; **AE_{glaucoma-}** = negative adversarial example for referable glaucoma; **AE_{glaucoma+}** = positive adversarial example for referable glaucoma; **AE_{ICDR-}** = negative adversarial example for increased cup-to-disc ratio; **AE_{ICDR+}** = positive adversarial example for increased cup-to-disc ratio; **AE_{RNFLD-}** = negative adversarial example for retinal nerve fiber layer defect; **AE_{RNFLD+}** = positive adversarial example for retinal nerve fiber layer defect; **AUC** = area under the receiver operating characteristic curve; **DLM** = deep learning model; **DRN** = disc rim narrowing; **FOV** = field of view; **GradCAM** = gradient-weighted class activation mapping; **ICDR** = increased cup-to-disc ratio; **RG** = referable glaucoma; **RNFLD** = retinal nerve fiber layer defect.

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